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Deterministic weather forecasting models based on intelligent predictors: A survey

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ABSTRACT

Weather forecasting is the practice of predicting the state of the atmosphere for a given location based on different weather parameters. Weather forecasts are made by gathering data about the current state of the atmosphere. Accurate weather forecasting has proven to be a challenging task for meteorologists and researchers. Weather information is essential in every facet of life like agriculture, tourism, airport system, mining industry, and power generation. Weather forecasting has now entered the era of Big Data due to the advancement of climate observing systems like satellite meteorological observation and also because of the fast boom in the volume of weather data. So, the traditional computational intelligence models are not adequate to predict the weather accurately. Hence, deep learning-based techniques are employed to process massive datasets that can learn and make predictions more effectively based on past data. The effective implementation of deep learning in various domains has motivated its use in weather forecasting and is a significant development for the weather industry. This paper provides a thorough review of different weather forecasting approaches, along with some publicly available datasets. This paper delivers a precise classification of weather forecasting models and discusses potential future research directions in this area.

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Contents

1. Introduction	3394
2. Related surveys	3395
3. Motivation	3395
4. Weather forecasting framework	3395
5. Classification of weather forecasting models	3396
5.1. Classification based on the methodology used	3396
5.1.1. Statistical models	3396
5.1.2. Artificial intelligence models	3397
5.1.3. Hybrid models	3403
5.2. Classification based on parameter to be predicted	3405
5.3. Discussion	3405

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6.	Results and analysis	3406
6.1.	Open weather datasets	3406
6.2.	Evaluation criteria	3407
6.3.	Analysis of forecasting models	3408
7.	Challenges and future directions	3409
8.	Conclusion	3410
	Declaration of Competing Interest	3410
	References	3410

Nomenclature

List of abbreviations

ACO	Ant colony optimization	KF	Kalman filter
AI	Artificial intelligence	LSSVM	Least square SVM
ANFIS	Adaptive neuro-fuzzy inference system	LSTM	Long short term memory
ANN	Artificial neural networks	MAE	Mean absolute error
API	Application programming interface	MAPE	Mean absolute percentage error
ARIMA	Autoregressive integrated moving average	ML	Machine learning
ARMA	Autoregressive moving average	MLP	Multi-layer perceptron
BPNN	Backpropagation neural network	MMO	Multi-objective optimization
CNN	Convolutional neural networks	MMODA	Modified multi-objective dragonfly algorithm
CRBM	Conditional restricted Boltzmann machine	MPR	Multi-variable polynomial regression
DBN	Deep belief network	MSE	Mean squared error
DBNPF	Deep belief network for precipitation forecast	NARX	Nonlinear autoregressive exogenous
DNN	Deep neural network	NMAE	Normalized mean absolute error
ELM	Extreme learning machine	NMSE	Normalized mean squared error
EMD	Empirical mode decomposition	NOAA	National oceanic and atmospheric administration
ENN	Elman neural network	OSORELM	Online sequential outlier robust ELM
f-ARIMA	Fractional-ARIMA	PSO	Particle swarm optimization
GARCH	Generalized AutoRegressive Conditional Heteroskedasticity	RBF	Radial basis function
GLM	Generalized linear model	RF	Random forests
GRNN	General regression neural network	RMSE	Root mean squared error
GRU	Gated recurrent unit	RNN	Recurrent neural networks
GSOD	Global surface summary of the day	SAE	Stacked autoencoders
GSR	Global solar radiation	SDAE	Stacked denoising autoencoders
HDT	Hybrid decomposition technique	SSA	Singular spectrum analysis
HMD	Hybrid mode decomposition	SVM	Support vector machine
IPSO	Improved PSO	SVR	Support vector regression
ISH	Integrated surface hourly	USAF	United states air force
		VAR	Vector autoregression
		WPD	Wavelet packet decomposition

1. Introduction

Weather forecasting is the scientific prediction of the state of atmospheric conditions such as temperature, humidity, dew point, rainfall, and wind speed based on reliable data. Barometers, radar, and thermometers are used to collect data for weather prediction. External factors such as current weather conditions, data of previous weather patterns, tracking the motion of air and clouds in the sky, finding and verifying changes in air pressure are essential in forecasting weather. Accurate weather forecasting information is crucial in keeping people and property safe from threats of tornadoes, floods, and major storms. Various sectors like Business, Tourism, Sports, Agriculture, Mining, Power, Food Industry, Airport, and Naval systems depend heavily on accurate weather forecasting. In agriculture, prior information about weather helps farmers to take necessary decisions to improve their crop yields. Airport or naval systems require continuous weather data to know if there is a sudden change in climatic conditions. Accurate wind speed prediction is critical for wind farms to control the working of wind turbines during wind power generation. Mining industries require precise weather information to monitor the Earth's crust continually.

Weather forecasting on a daily, weekly, monthly, or yearly basis becomes essential as it can better reflect the changing trend of climate and also provide timely and efficient environmental information for the decisions at the micro-management level. Moreover, in specific cases, accurate weather prediction helps to prevent the occurrence of floods or droughts (Salman et al., 2015).

The rapid development of technologies like the Internet of Things, Wireless Sensor Network, and Cloud Computing has helped weather forecasting enter the era of Big Data. Big data technologies help to predict the future climate states more precisely. Moreover, with the advancement of deep learning techniques and appropriate data visualization methods, weather forecasting, and climate prediction can be made more effectively and accurately. Hence it is reasonable to use deep learning approaches to mine valuable information from weather data. Deep learning techniques use layers of neural networks to identify and extract meaningful patterns from the datasets. A neural network with deep architectures can extract high-level abstract features of Big Data accurately (Gheisari et al., 2017). In order to predict the weather in a very effective way, several weather forecasting models using deep learning have been proposed. The foremost intention of this paper

is to provide an extensive review of weather forecasting models and various techniques and methodologies currently used by various researchers for weather forecasting. The main highlights of the paper are presented below.

- (1) A comprehensive review of various weather forecasting models
- (2) Analysis of the essential hyper-parameters used by the forecasting models
- (3) A precise classification of various weather forecasting models
- (4) Exploration of the effectiveness of the various models using statistical error indicators
- (5) Serve as a guide for beginners who are passionate about research on weather forecasting and acquire knowledge on different techniques and available Open datasets.
- (6) This review deliberates potential future research directions in this area.

This paper is organized as follows. [Section 2](#) gives an overview of related surveys, and [section 3](#) explains the motivation behind the proposed work. [Section 4](#) describes the proposed framework, where the general weather forecasting framework is presented, followed by the classification of weather forecasting models in [section 5](#). [Section 6](#) investigates the results, and [section 7](#) gives a brief note on challenges and future directions. Finally, [Section 8](#) concludes the paper.

2. Related surveys

In this section, some of the surveys related to weather forecasting are reviewed. Gheisari et al. (2017) performed a study on big data and its research directions and analyzed various deep learning architectures suitable for big data processing, its challenges, and future trends. Saima et al. (2011) surveyed various weather forecasting models, including statistical, artificial intelligence, and hybrid models, and analyzed their merits and demerits. Nagaraja et al. (2016) reviewed forecasting models for wind energy and attempted to find reliable forecasting models to explore the behavior of wind. Liu et al. (2019) focused on wind speed forecasting models using machine learning predictors, deep learning predictors, and hybrid models. The main machine learning predictors used for weather forecasting are Artificial Neural Networks, Support Vector Machines, and Extreme Learning Machines. Deep learning predictors consist of Autoencoders, Recurrent Neural Networks, and Long Short Term Memory Networks. Time series forecasting becomes possible using these methods.

Kunjumon et al. (2018) performed a study on various forecasting models based on artificial neural networks, support vector machines, and decision trees. Naveen and Mohan (2019) discussed various application domains of weather forecasting and also investigated weather prediction models using machine learning techniques and the challenges related to weather forecasting. Reddy and Babu (2017) analyzed various big data weather prediction models using MapReduce and machine learning models. The authors also addressed the limitations and issues related to big data weather forecasting, especially rainfall forecasting. Tran-Anh et al. (2019) recommended an improved rainfall prediction model based on wavelet transform and seasonal ANN. The authors also investigated different methods for predicting monthly rainfall. Tarade and Katti (2011) offered a comprehensive study of wind speed forecasting systems based on ARIMA, ANN, and polynomial curve fitting. Kulkarni et al. (2008) investigated the effectiveness of various statistical approaches suitable for wind speed forecasting.

Experimental results verified that extrapolation using periodic curve fitting and ANN are better models for wind speed prediction than ARIMA and produced reliable results. Medina et al. (2019) made a comparative study of demand forecasting models using ANN and observed that neural network produced more accurate results compared to other methods. Rasp and Lerch (2018) studied the impact of employing neural networks for post-processing of ensemble weather prediction models to determine the temperature in Germany. Siami-Namini et al. (2018) conducted an extensive survey on various time series prediction models using LSTM and ARIMA. Hong et al. (2020) performed a comprehensive review of energy forecasting models and summarized current research trends along with different publicly available datasets for energy forecasting research. The proposed framework provides a precise classification of various weather forecasting models. The survey also delivers the state of the art models for weather forecasting, its challenges, and future directions.

The proposed framework is a different approach for surveying the existing weather forecasting models, where weather forecasting models are classified mainly based on the methodology employed. The survey also classifies the models based on the weather parameter to be predicted and provides the details of the open datasets available for experiments. The following section details the motivation behind the survey.

3. Motivation

The weather forecasting systems can be classified into three categories based on the model or methodology employed for forecasting, namely statistical models, Artificial Intelligence models, and hybrid models. Among the three categories, statistical models deal with linear datasets. The commonly applied statistical models include ARMA, ARIMA, and its variants. Artificial Intelligence models are further classified into machine learning predictors and deep learning predictors. Machine learning and deep learning models are better for handling nonlinear datasets. Artificial Neural Networks, Support Vector Machine, Extreme Learning Machine, Autoencoders, Convolutional Neural Networks, Recurrent Neural Networks, and Long Short Term Memory Networks are some of the popular AI models employed for weather forecasting tasks. Hybrid models are combinations of two or more models that aim to enhance the performance of the forecasting models further. The main aim of weather forecasting models is to predict temperature, humidity, dew point, rainfall, and wind speed based on past data. Analysis of the literature reveals that several surveys related to wind speed forecasting have been conducted. However, reliable and comprehensive surveys related to forecasting weather parameters, such as temperature, rainfall, dew point, and humidity, are much less. Hence, an extensive survey on various weather forecasting models and their classifications are presented in this paper.

4. Weather forecasting framework

The general framework of a weather prediction model is described in [Fig. 1](#). The various steps of the weather prediction framework include data acquisition, data pre-processing, model selection and training, model evaluation, and visualization of results. Due to the rapid development of novel technologies, such as the Internet of Things, Wireless Sensor Networks, and Cloud Computing, weather data were made available in various formats and huge volumes (Chavan and Momin, 2017). This data consists of enormous amounts of both useful and useless information, and they are also in the form of unstructured data. Once the data is gathered, the subsequent step is to pre-process the data to

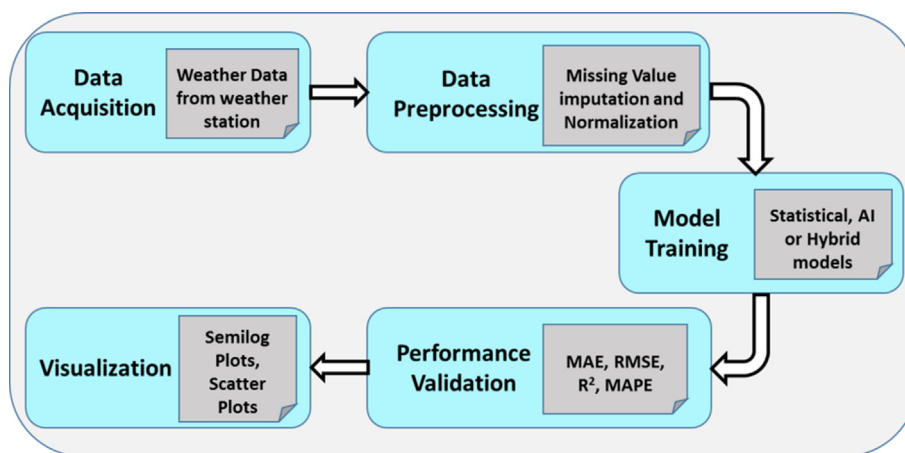


Fig. 1. The layout of the weather forecasting model.

remove irrelevant and missing values and to get cleaned data. The primary forms of data pre-processing include data cleaning, data integration, data reduction, and data transformation. Data pre-processing techniques aim to improve the quality of the input data. Quality data will always yield quality output during the training process. Therefore, at this stage, data is prepared for training. Data cleaning methods remove noise, missing values, and inconsistencies present in the data. Real-world data often contain missing values. It can be instigated by issues such as values that are not documented and data corruption. Most of the forecasting algorithms do not support data with missing values. So, missing value imputation is a necessity, and the missing values are imputed in most cases using mean values.

The data integration process merges data from different sources into an intelligible data store. Data reduction can reduce the size of the data by aggregating, eliminating redundant features, or clustering. In the data transformation step, the data are normalized using the normalization or standardization technique and converted to a form suitable for processing. Data transformation techniques not only improve the accuracy and efficiency of mining algorithms but also reduces the training time of the models. These pre-processing techniques are not mutually exclusive, and hence, they may function together. Once the data is pre-processed, a suitable forecasting model is selected, trained, and tested using the datasets. Appropriate algorithms can be used for effective forecasting of weather information.

Model selection and training is an essential step in any forecasting system. The knowledge about the various types of forecasting models will help the researchers in choosing a suitable forecasting model. A suitable forecasting model matching the application domain is selected and trained using the training datasets. Upon completion of the training, the performance of the model is evaluated using statistical quantitative error indicators such as MAE, RMSE, MAPE, and R^2 . Lastly, the results are visualized using suitable plots. Scatter plots, line plots, and semilog plots can be employed to visualize the results. These plots can be utilized to analyze graphically, the difference between actual and predicted values. The variation of predicted values from actual values can be visualized more precisely using semilog plots.

5. Classification of weather forecasting models

The weather forecasting systems are classified into various categories based on multiple factors. The factors comprise the number of variables involved, the number of time steps predicted, the method-

ology used, the prediction horizon, and the parameter to be predicted. Fig. 2 provides the complete classification of the forecasting models based on various factors mentioned above. The forecasting models can be univariate or multivariate depending upon the number of variables involved in forecasting. For instance, if the temperature is predicted based on many environmental factors, then such type of forecasting can be termed as multivariate. A univariate model depends on only one variable, whereas a multivariate model predicts the output based on several factors. Apart from this, forecasting models can be single-step or multistep, depending on the number of time steps to be predicted. Single-step forecasting models predict a single observation in the future, whereas multistep forecasting models predict multiple time steps to the future. Weather forecasting approaches come in four different scales based on the time horizon: very short-term, short-term, medium-term, and long term. Short-range and very-short range forecasts are found to be more accurate than medium or long ranges.

Furthermore, the forecasting systems can be two basic types namely, deterministic and probabilistic, based on the methodology employed for forecasting. Deterministic methods provide accurate values of weather forecasts for a specific location, while probabilistic methods suggest probabilities of weather events. This survey basically focuses on deterministic forecasting models. The deterministic forecasting models can be further classified as statistical models, Artificial Intelligence models, and hybrid models. Depending on the parameter to be predicted, forecasting models can be classified as temperature prediction models, wind speed prediction models, rainfall prediction models, dew point prediction models, etc. as illustrated in Fig. 2.

With the influence of weather in social and economic activities, adverse events can be prevented by predicting weather conditions accurately. In the preceding years, several methods have been suggested to deal with this goal. The next section elucidates the different categories of weather forecasting models.

5.1. Classification based on the methodology used

Based on the methodology used for forecasting, deterministic models can be classified as statistical models, Artificial Intelligence models, and hybrid models.

5.1.1. Statistical models

Statistical forecasting models are linear models and can be employed for very short-term weather forecasting. Statistical forecasting allows weather predictions using past weather data. Linear

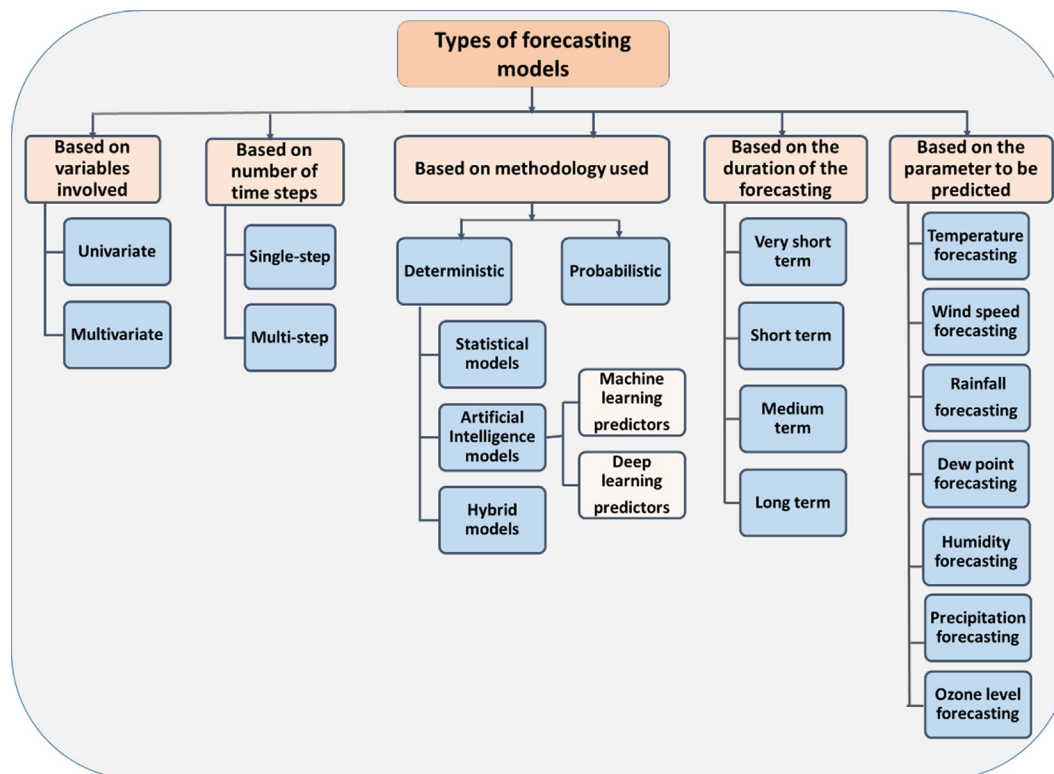


Fig. 2. Classification of forecasting models.

regression is a statistical model and is used for finding the linear relationship between study variables and one or more explanatory or independent variables. Linear regression can be of two types, namely simple and multiple. Simple linear regression is convenient for finding the statistical relationship between two continuous variables, and the aim is to obtain a line that best fits the data for which total prediction error is as small as possible. A gradient descent optimization algorithm is employed to minimize the error as small as possible. Multiple linear regression differs from simple linear regression in such a way that the study argument depends on more than one independent arguments. ARMA, ARIMA, multiple regression, and VAR are some of the commonly used statistical models. ARIMA is considered a statistical model that uses time-series data to predict future observations. Vector autoregression is also a statistical model used for forecasting a vector of time series. VAR model is the generalized multivariate form of the univariate autoregression model. This model is beneficial when predicting a collection of related variables. Tarade and Katti (2011) and Kulkarni et al. (2008) studied wind speed forecasting systems based on various statistical approaches. Khashei and Bijari (2011) proposed a model by combining linear model ARIMA and nonlinear model ANN together and found them more effective than traditional individual models. It can be employed as a hybrid model in the time series forecasting field, especially when higher forecasting accuracy is desired. Zaw and Naing (2009) suggested a rainfall prediction model using multi-variable polynomial regression (MPR) and compared the performance of the model with a multiple linear regression model. Kavasseri and Seetharaman (2009) presented a wind speed and power prediction model using the fractional-ARIMA for North Dakota, USA, and compared the performance of the forecasting model with the persistence model. Erdem and Shi (2011) proposed wind speed prediction models based on ARMA, decomposed ARMA, VAR, and restricted VAR and investigated the performance of these models on two datasets collected from wind farms in North Dakota.

Liu et al. (2011) proposed a predictive model for wind speed using ARMA–GARCH method with 7-year hourly data collected from Colorado, USA. Liu et al. (2012) suggested wind speed forecasting models by combining ARIMA with ANN and ARIMA with Kalman separately and analyzed the effectiveness of these models in multi-step prediction. Chen et al. (2009) modeled a system to forecast wind power using ARIMA. A forecasting system using ANFIS and ARIMA was proposed by Suhartono et al. (2012) for predicting the monthly rainfall of Indonesia. Cadenas et al. (2016) recommended a wind speed forecasting system with the help of a univariate ARIMA and a multivariate NARX model. Table 1 provides the parameters and the metrics used by the statistical models discussed in the literature.

5.1.2. Artificial intelligence models

The development of Artificial Intelligence has inspired the growth of intelligent forecasting models. These models are proved to be robust and effective compared to statistical models. AI models are capable of dealing with non-linear datasets effectively and exhibit better forecasting performance. These models are further categorized into machine learning predictors and deep learning predictors, as shown in Fig. 2.

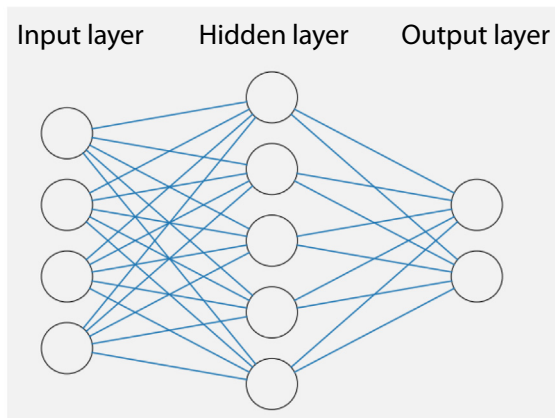
5.1.2.1. Machine learning predictors. Machine learning and deep learning models are better models for handling nonlinear datasets. ANN, SVM, ELM, and Random Forests are some of the popular machine learning predictors used for weather forecasting. In this section, the various weather prediction systems using machine learning techniques are presented. They include ANN-based models, SVM-based models, and other ML models.

5.1.2.1.1. ANN-based models. Neural networks based models, which are supervised and predictive, are considered to be one of the most popular techniques for weather prediction because they can capture non-linear relationships of past weather trends and future weather conditions. Neural networks function as nonlinear

Table 1

Summary of the parameters used by statistical models.

Algorithm Used	Reference	Parameters used									Metrics used				
		Temperature	Wind speed	Wind direction	Air pressure	Humidity	Precipitation	Rainfall	Dew point	Ozone	R ²	MSE	RMSE	MAE	MAPE
ARIMA	(Tarade and Katti, 2011)		✓								✓				
ARIMA	(Kulkarni et al., 2008)		✓										✓		
MPR	(Zaw and Naing, 2009)							✓					✓		
f-ARIMA	(Kavasseri and Seetharaman, 2009)		✓										✓		
ARMA, VAR	(Erdem and Shi, 2011)		✓											✓	
ARMA-GARCH	(Liu et al., 2011)		✓								✓				
ARIMA-KF	(Liu et al., 2012)		✓									✓		✓	✓
ARIMA	(Chen et al., 2009)		✓									✓			
ARIMA, ANFIS	(Suhartono et al., 2012)							✓					✓		
ARIMA, NARX	(Cadenas et al., 2016)		✓									✓		✓	

**Fig. 3.** Structure of a neural network.

regression models with rapid information processing capability and develop a mapping of the input and output variables.

An Artificial Neural Network comprises a network of neurons termed nodes, connected to each other. Within each node, a transfer function is employed to convert the inputs to the output. Fig. 3 shows the architecture of a single hidden layer neural network. In general, a neural network system comprises an input, output, and hidden layers. The nodes within the input layer take in information, transform this information using activation functions, and passes it to the next node. Each node aggregates the value it obtains and then transforms the results depending upon its activation function. These transformed values flow from the input layer through hidden layers until the output nodes are reached. The association between the input X and output Y of the neural network at layer i can be represented as in equation (1).

$$Y_i = g\left(\sum_{j=1}^n W_{ij}X_j + b_i\right) \quad (1)$$

where W_{ij} represents the weight associated with the connection between neurons, b_i signifies the bias of the neuron i and g denotes

the activation function. The difference between the predicted value and the actual value is calculated, and this error is propagated backward by adjusting the weights to minimize error using optimization algorithms like Gradient Descent. Optimization algorithms are employed as they are essential in building efficient neural networks.

In this subsection, the various Weather prediction models developed using Artificial Neural Networks proposed by different authors are presented. Nayak et al. (2012) proposed a neural network model for weather forecasting and compared the results with SVM and Fuzzy Logic. Liu et al. (2018) presented a temperature forecasting model based on the neural network approach and studied the importance of neural networks in intrusion detection. Hung et al. (2009) proposed an ANN-based real-time rainfall forecasting model for managing flood in Bangkok, Thailand, and the proposed model showed better forecasting accuracy than the persistent model. Kashiwao et al. (2017) observed that ANN gives more accurate results to predict the probability of rainfall.

Abhishek et al. (2012) proposed a valid and reliable nonlinear predictive model using Artificial Neural Networks for daily weather forecasting and compared the performance of the proposed model using different activation functions, hidden layers, and neurons to predict maximum temperature. Tests are conducted by varying the number of neurons and hidden layers and concludes that performance increases with the number of hidden layers. A two hidden layer Back Propagation Neural Network algorithm for accurate rainfall prediction in Tenggarong, East Kalimantan - Indonesia is proposed by Mislani et al. (2015). The metric mean square error is employed to measure the performance of the prediction task. Khajure and Mohod (2016) proposed a future weather prediction model using neural networks combined with the Fuzzy Inference System. Neural Networks is trained using different combinations of weather parameters such as humidity, temperature, pressure, wind speed, dew point, and visibility, and then predicted values are applied to the Fuzzy Inference System. The combination of a neuro-fuzzy system is used to enhance the accuracy of prediction.

Suksri and Kimpan (2016) suggested a weather forecasting method using Artificial Neural Networks to predict daily mean

temperature, where the Fireworks Algorithm is employed to optimize the parameters of ANN. Fireworks Algorithm is a novel Swarm Intelligence algorithm for optimization. Narvekar and Fargose (2015) presented a weather forecasting model using ANN with a backpropagation algorithm, where the different input parameters taken into consideration are temperature, relative humidity, air pressure, wind speed and direction, cloud amount, and rainfall. The efficiency of the model is assessed by determining the mean squared error. Akbar et al. (2018) suggested an ANN-based model for predicting oil yield in turmeric, and the results confirmed that the proposed neural network model has better performance. Wollsen and Jorgensen (2015) presented a novel method to forecast future weather using NARX (Nonlinear Autoregressive Exogenous) and ANN models. Filik and Filik (2017) described an ANN-based model to predict wind speed using weather parameters such as wind speed, temperature, and pressure values and investigated the effect of these parameters. Sobrevilla (2016) developed an ANN-based model for daily weather prediction, and also analyzed the importance of missing value imputation in the datasets. It has been perceived that the proposed model produced reasonable accuracy.

5.1.2.1.2. SVM-based models. SVM is a Supervised Machine Learning algorithm that can solve both classification and regression tasks (Awad and Khanna, 2015; Kavitha et al., 2016). SVM is formally defined by a hyperplane that has the highest minimum distance to the training data samples. An optimal hyperplane is a hyperplane that has the maximum margin of training data. The neighboring training samples to the hyperplane are termed as Support Vectors. Support Vector machines can find nonlinear solutions efficiently using the “Kernel trick,” and they also work well with high-dimensional datasets. Kernels are mathematical functions that convert inputs into the necessary format. The various kernel functions employed by SVM algorithms can be linear, nonlinear, polynomial, radial basis function, and sigmoid.

Let (x_j, y_j) be training datasets where j varies from 1 to N and N indicates the number of samples. The SVR algorithm computes output y from input data x using the equation (2).

$$y = f(x) = w^T \phi(x) + b \quad (2)$$

where $\phi()$ indicates the nonlinear mapping function, which maps input data to a high dimensional feature space. The variables w and b are the weight vector and bias, respectively. The aim of ϵ -SVR is to find a function $f(x)$ that has at most ϵ deviation from the targets y_j for all the training data. This problem can be defined as convex optimization function as in Equation (3).

$$\text{minimize } \frac{1}{2} \|w\|^2 \quad \text{subject to } \begin{cases} y_j - w^T \phi(x_j) - b \leq \epsilon \\ w^T \phi(x_j) + b - y_j \leq \epsilon \end{cases} \quad (3)$$

The above-defined convex optimization problem is feasible when function $f()$ approximates all training pairs (x_j, y_j) with ϵ precision. However, to deal with infeasible constraints, two slack variables ξ_j and ξ_j^* are presented. Thus, equation (3) can be extended by including slack variables as shown in equation (4).

$$\text{minimize } \frac{1}{2} \|w\|^2 + C \sum_{j=1}^N (\xi_j + \xi_j^*) \quad \text{subject to} \quad \begin{cases} y_j - w^T \phi(x_j) - b \leq \epsilon + \xi_j \\ w^T \phi(x_j) + b - y_j \leq \epsilon + \xi_j^* \\ \xi_j, \xi_j^* \geq 0 \end{cases} \quad (4)$$

The variables C and ϵ are predefined parameters and should be greater than 0. C is the regularization parameter. The above regression formulation deals with ϵ -intensive loss function $|\xi|_\epsilon$ and is defined as in equation (5).

$$|\xi|_\epsilon = \begin{cases} 0 & \text{if } |\xi| \leq \epsilon \\ |\xi| - \epsilon & \text{otherwise} \end{cases} \quad (5)$$

The dual design permits the extension of SVR to nonlinear functions. The optimization problem can be further extended by using the Lagrangian form as in equation (6).

$$L = \frac{1}{2} \|w\|^2 + C \sum_{j=1}^N (\xi_j + \xi_j^*) - \sum_{j=1}^N \alpha_j (\epsilon + \xi_j - y_j + w^T \phi(x_j) + b) - \sum_{j=1}^N \alpha_j^* (\epsilon + \xi_j^* + y_j - w^T \phi(x_j) - b) - \sum_{j=1}^N (\eta_j \xi_j + \eta_j^* \xi_j^*) \quad (6)$$

The variables α_i, α_i^* are Lagrange multipliers. It is also assumed that $\alpha_j, \alpha_j^*, \eta_j, \eta_j^* \geq 0$. Equation (6) can be reduced by taking the partial derivatives of L with respect to the variables (b, w, ξ_j, ξ_j^*) and equated to zero based on Karush-Kuhn-Tucker (KKT) conditions (Smola and Schölkopf, 2004) as in equations (7)–(10).

$$\frac{\partial L}{\partial b} = \sum_{j=1}^N (\alpha_j^* - \alpha_j) = 0 \quad (7)$$

$$\frac{\partial L}{\partial w} = w - \sum_{j=1}^N (\alpha_j - \alpha_j^*) \phi(x_j) = 0 \quad (8)$$

$$\frac{\partial L}{\partial \xi_j} = C - \alpha_j - \eta_j = 0 \quad (9)$$

$$\frac{\partial L}{\partial \xi_j^*} = C - \alpha_j^* - \eta_j^* = 0 \quad (10)$$

Equation (6) can be rewritten by substituting the values of equations (7)–(10) as given below.

$$\text{maximize } \left\{ -\frac{1}{2} \sum_{i,j=1}^N (\alpha_i - \alpha_i^*) (\alpha_j - \alpha_j^*) \phi(x_i)^T \phi(x_j) - \epsilon \sum_{i=1}^N (\alpha_i + \alpha_i^*) + \sum_{i=1}^L y_i (\alpha_i - \alpha_i^*) \right\} \quad \text{such that } \sum_{i=1}^N (\alpha_i - \alpha_i^*) = 0 \text{ and } \alpha_i, \alpha_i^* \in [0, C] \quad (11)$$

The dual variables η_j, η_j^* are eliminated from equation (11) using the conditions stated in equations (9) and (10). From equation (8), the parameter w can be defined as a nonlinear combination of the training samples y_i as shown in equation (12).

$$w = \sum_{i=1}^N (\alpha_i - \alpha_i^*) \phi(x_i) \quad (12)$$

The target function $f(x)$ can then be defined using equation (13).

$$f(x) = \sum_{i=1}^N (\alpha_i - \alpha_i^*) \phi(x_i)^T \phi(x) + b \quad (13)$$

The kernel function $k(x_i, x)$ can be defined as a linear dot product of the nonlinear mapping as in equation (14).

$$k(x_i, x) = \phi(x_i)^T \phi(x) \quad (14)$$

hence, target function $f(x)$ given in equation (13) can be rewritten as in equation (15).

$$f(x) = \sum_{i=1}^N (\alpha_i - \alpha_i^*) k(x_i, x) + b \quad (15)$$

In this subsection, the various weather prediction models developed using SVM are presented. SVM exhibits satisfactory accuracy under a limited number of training samples. So, Du et al. (2017) recommended an SVM based precipitation forecasting model, where the Particle Swarm Optimization (PSO) algorithm is employed to optimize the hyperparameters of the SVM model.

Yu et al. (2016) suggested a novel temperature forecasting model based on Least Square SVM (LSSVM), where the IPSO algorithm optimizes parameters of LSSVM. The authors compared the performance of IPSO-LSSVM with SVM and ANN models. Rasel et al. (2017) recommended a weather forecasting model using SVM and ANN to predict the weather of Chittagong, Bangladesh, and the results showed that SVM outperformed ANN in rainfall prediction. Lu and Wang (2011) proposed a model for rainfall prediction using a support vector machine to forecast monthly rainfall in China. According to the authors, the accuracy of the model is proved to be very promising and encouraging. Shabariram et al. (2016) developed a tool for predicting rainstorm from a large amount of rainfall data in an efficient manner using Support Vector Machine.

5.1.2.1.3. Other ML models. Random forest is one of the reliable ensemble classifiers for high dimensional data and is proved to be better for weather prediction. Singh et al. (2019) offered a weather prediction system using a random forest algorithm. A model for forecasting the likelihood of rain, average temperature, and maximum wind for a given day is developed by Ahmed (2015). The authors employed the classifiers such as Naïve Bayes, Simple Random Forests, J48, and IB1 for the experiments. Hasan et al. (2016) proposed a forecasting model where C4.5 learning algorithm is utilized for predicting weather events. Stern (2008) offered a medium-range weather forecasting model for Melbourne, Australia. A dew point temperature prediction model based on an extreme learning machine was presented by Mohammadi et al. (2015). The effectiveness of the suggested model is compared with the SVM and ANN models. The simulation results verified that the ELM based proposed model outpaced both SVM and ANN models. The error metrics used to measure the accuracy are Mean Absolute Bias Error, Root Mean Square Error, and correlation coefficient R. Kurniawan et al. (2017) recommended a weather prediction system based on the fuzzy logic algorithm. A novel method for plant watering based on weather prediction is also introduced in this paper. The proposed weather calculation system will help the General Farming Automation Control System to work automatically. Juneja and Das (2019) proposed methods for enhancing the quality of datasets by applying pre-processing techniques.

Weather prediction models based on big data technologies are described below. In the big data era, traditional methods of data processing have been replaced by big data technologies. Ismail et al. (2016) proposed an analytical big data framework for temperature forecasting based on the MapReduce algorithm. Radhika et al. (2017) presented an outline of a few strategies in supporting big data administration and investigation in the geoscience domain for climate studies. The paper also discussed an overview of HBase for storing and managing essential geoscience data across distributed machinery. There are various techniques to manage big data from the geospatial domain, and these techniques include Data fusion, Crowdsourcing, Cluster analysis, and Machine Learning. Jayanthi and Sumathi (2017) and Adam et al. (2017) developed big data frameworks for weather forecasting using the MapReduce algorithm and Spark, respectively. Suryanarayana et al. (2019) built a weather forecasting system using big data technology Hadoop MapReduce. Pandey et al. (2017) investigated the application of big data in the field of weather prediction, and a Hadoop based weather forecasting model is proposed for efficient processing and prediction of weather data. The word count algorithm is used to pre-process the weather data. Two data mining tools, fuzzy logic and ANFIS methods, are applied for accurate prediction of weather data. Experimental results showed that the ANFIS method gives more accurate results compared to other methods. Various application areas that could benefit from weather forecasting using

big data analytics are described by Himanshi and Raksha (2017), and also the challenges faced while forecasting weather are explained. Navadia et al. (2017) proposed a predictive analysis model using Apache PIG on Hadoop to forecast the chances of rainfall. The proposed system serves as a tool that takes in the rainfall data as input and efficiently predicts future rainfall. Table 2 summarizes the parameters and the metrics used by the machine learning predictors.

5.1.2.2. Deep learning predictors. Deep Learning is a subclass of machine learning which uses an artificial neural network approach for getting intelligence from massive datasets, and it mainly makes use of supervised or unsupervised strategies in deep architectures to learn hierarchical representations automatically. Big data can be more accurately processed using deep learning techniques. Deep learning models have been successfully implemented in many areas such as object detection, speech recognition, visual object recognition, genomics, and time series prediction. Deep learning techniques are more reliable for time series prediction because of their ability to learn temporal dependence present in time series data. Deep learning uses neural networks with deep architectures, which are composed of many layers of non-linear processing stages. The output of each layer is given as input to its next higher layer. Fig. 4 shows the architecture of a deep neural network with four hidden layers. Feed-forward neural networks or MLPs with many hidden layers are indeed an excellent example of the models with a deep architecture. Deep learning is a vigorous area of research today, and Microsoft, Google, IBM, and Facebook are doing active research in this area (Deng, 2014). Autoencoders and recurrent neural networks are the popular deep learning architectures employed for time series data.

5.1.2.2.1. Autoencoders. An autoencoder is a neural network-based unsupervised machine learning algorithm. Furthermore, it reconstructs the output values the same as the inputs. An autoencoder has three layers, the input layer, the hidden layers, and the output layer. The hidden layer encodes the inputs to obtain a hidden representation, and the output layer decodes the hidden representation. The network is trained in such a way to reconstruct its input to its output by learning abstract representations of the inputs. Since Autoencoders reconstruct their input, the dimensionality of the input and the output layers must be equal.

Autoencoders are a form of feed-forward neural network and have two components, namely encoder, and decoder. The encoder part performs encoding of input x_i to a hidden pattern h as given in equation (16). The decoder part decodes the input from the hidden pattern as in equation (17). The loss function used by Autoencoders is squared error loss function and is defined in equation (18). The weights, bias, and reconstructed input are denoted by W , b , and $\hat{x}_i$, respectively. The activations or transfer functions are indicated using g and f . Backpropagation algorithm is employed to train the networks. The main applications of Autoencoders include dimensionality reduction and denoising of data (Goodfellow et al., 2016).

Autoencoders can be of two types, undercomplete and overcomplete autoencoders. Undercomplete autoencoders have a fewer number of neurons in the hidden layer than the input layer and are used for feature extraction and dimensionality reduction tasks. Whereas overcomplete autoencoders have a higher number of neurons in the hidden layer than the input layer. The architecture of a two-layer stacked undercomplete autoencoder is shown in Fig. 5. The Figure shows two parts, the encoder part encodes inputs to a hidden pattern, and the decoder part reconstructs the input from the hidden pattern. Autoencoders can represent both linear and nonlinear transformation.

$$h = g(Wx_i + b) \quad (16)$$

Table 2
Summary of the parameters used by Machine learning models.

Methodology	Algorithm Used	Reference	Parameters used									Metrics used				
			Temperature	Wind speed	Wind direction	Air pressure	Humidity	Precipitation	Rainfall	Dew point	Ozone	R ²	MSE	RMSE	MAE	MAPE
ANN based models	ANN	(Tran Anh et al., 2019)								✓				✓	✓	✓
	ANN	(Tarade and Katti, 2011)		✓									✓			
	ANN	(Kulkarni et al., 2008)		✓										✓		
	ANN	(Nayak et al., 2012)	✓	✓			✓			✓						✓
	ANN	(Liu et al., 2018)	✓									✓	✓			
	ANN	(Hung et al., 2009)							✓			✓		✓		
	ANN	(Kashiwao et al., 2017)							✓			✓				
	ANN	(Abhishek et al., 2012)	✓	✓					✓				✓			
	ANN	(Mislan et al., 2015)							✓				✓			
	ANN	(Khajure and Mohod, 2016)	✓	✓		✓	✓			✓						
	Fireworks - ANN	(Suksri and Kimpan, 2016)	✓	✓	✓		✓	✓	✓	✓				✓		
	ANN	(Narvekar and Fargose, 2015)	✓	✓	✓	✓	✓		✓	✓			✓			
ANN	(Filik and Filik, 2017)	✓	✓		✓								✓		✓	
SVM based models	PSO - SVM	(Du et al., 2017)						✓					✓			
	SVM	(Yu et al., 2016)	✓													
	SVM	(Rasel et al., 2017)							✓					✓		✓
	SVM	(Lu and Wang, 2011)							✓				✓			✓
	SVM	(Shabariram et al., 2016)							✓							
Other ML models	RF	(Singh et al., 2019)	✓						✓			✓				
	RF	(Ahmed, 2015)	✓	✓					✓			✓				
	C4.5	(Hasan et al., 2016)	✓	✓		✓	✓		✓	✓						
	ELM	(Mohammadi et al., 2015)									✓		✓			✓
	MapReduce	(Ismail et al., 2016)	✓													
	MapReduce and Spark	(Suryanarayana et al., 2019)	✓	✓												
	MapReduce	(Pandey et al., 2017)	✓				✓	✓					✓			
	Apache PIG	(Navadia et al., 2017)					✓	✓								

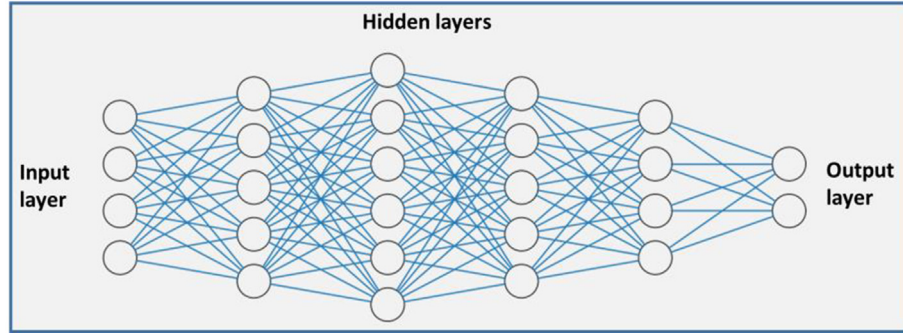


Fig. 4. Structure of a deep neural network.

$$\hat{x}_i = f(Wh + c) \quad (17)$$

$$J(\Theta) = \min \frac{1}{m} \sum_{i=1}^n \sum_{j=1}^n (\hat{x}_{ij} - x_{ij})^2 \quad (18)$$

Denoising Autoencoders, Stacked Autoencoders, Stacked Denoising Autoencoders, and Variational Autoencoders are some of the forms of autoencoders. The main goal of autoencoders is to set the output the same as the input. However, sometimes the reconstruction never ensures perfect results. This may be because of information loss during extraction. Denoising autoencoders reproduce pure input from the corrupted version of the input.

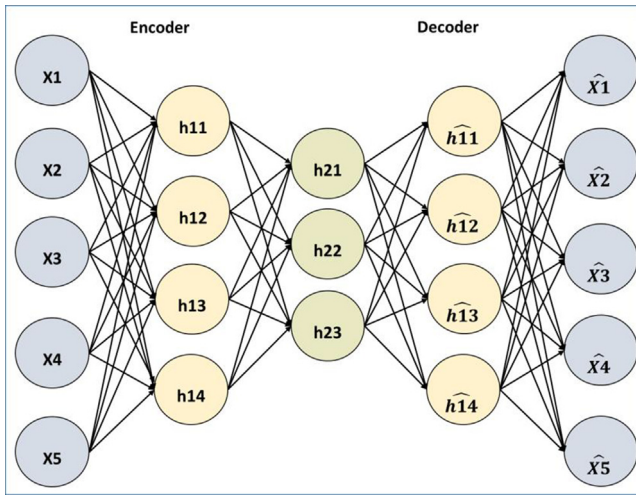


Fig. 5. The architecture of a stacked autoencoder.

Stacked autoencoders are produced by stacking multiple autoencoders in which the outputs of each layer are linked to the inputs of the succeeding layer. The training of stacked autoencoders is performed using a greedy layer-wise strategy. Stacked Denoising Autoencoders are a collection of denoising autoencoders stacked together. After pre-training all the layers, fine-tuning for the entire network is performed. Unlike ordinary autoencoders, variational autoencoders are generative models that take random numbers as inputs and generate images as output.

5.1.2.2.2. Recurrent neural networks. Deep learning architectures like RNN and LSTM are deep-rooted models for time series prediction. Recurrent Neural Networks are not appropriate for learning long term dependencies present in time series data and can be used to learn only short term dependencies precisely. This is because Recurrent Neural Networks suffer from vanishing and exploding gradients (Nielsen, 2015). This unstable gradient problem may occur when the gradient of the transfer function becomes too small or too large. This makes RNN hard to train and also prevents it from catching long term dependencies. These shortcomings can be rectified using LSTM networks, a variant form of Recurrent Neural Networks. LSTM based architecture is capable of capturing long term dependencies more precisely.

LSTM networks were familiarized by Hochreiter and Schmidhuber in 1997 (Hochreiter and Schmidhuber, 1997). Besides input and output layers, LSTM networks comprise one or more hidden layers. Each LSTM cell in the hidden layer consists of three gates: the input gate, output gate, and forget gate. The layout of an LSTM cell is depicted in Fig. 6. Forget gate attempts to increase the efficiency of the network by eliminating less important information from the cell state. The input gate transfers new information to the cell state, while the output gate passes valuable information from the memory cell to the output. The mathematical representation of an LSTM network is given in equations (19) to (24).

$$o_t = \sigma(W_o h_{t-1} + U_o x_t + b_o) \quad (19)$$

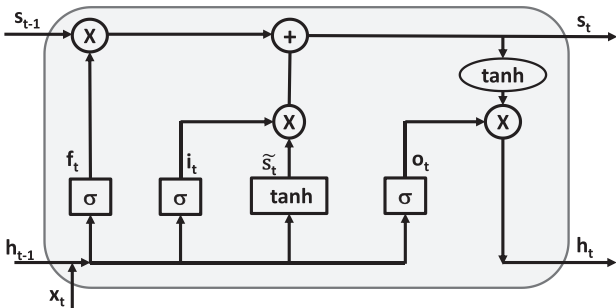


Fig. 6. Layout of LSTM cell.

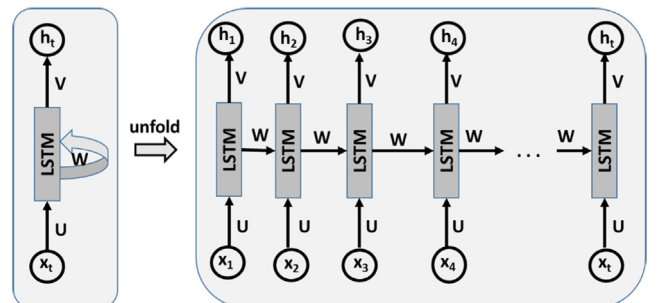


Fig. 7. The layout of an LSTM network.

$$i_t = \sigma(W_i h_{t-1} + U_i x_t + b_i) \quad (20)$$

$$f_t = \sigma(W_f h_{t-1} + U_f x_t + b_f) \quad (21)$$

$$\tilde{s}_t = \tanh(W h_{t-1} + U x_t + b) \quad (22)$$

$$s_t = f_t \odot s_{t-1} + i_t \odot \tilde{s}_t \quad (23)$$

$$h_t = o_t \odot \tanh(s_t) \quad (24)$$

where i_t , o_t , and f_t represent input gate, output gate, and forget gate, respectively. s_t , x_t , and h_t specify the state of the network, input, and output respectively at time step t . $\tilde{s}_t$ signifies a temporary state. U , V , and W denote the weights, and b is the bias. The operator $\odot$ symbolizes the element-wise multiplication.

Fig. 7 illustrates the structure of a single hidden layer LSTM network. Stacked or Deep LSTM consists of multiple layers, where the output of the previous layer is the input to the following layer. The notion of Stacked LSTM was first suggested by Graves et al. (2013). Stacked LSTM cells can store more information through hidden layers. Each LSTM layer processes a portion of the job and transfers the output to the next layer. Lastly, the results are taken from the output layer. The stacking of hidden layers deepens the Recurrent Model and can help learn features more correctly. Stacked LSTMs are a powerful technique for sequence prediction tasks. GRU is another form of RNN which can also be used for solving time series prediction problems.

In this subsection, the various weather prediction models developed using deep learning architectures are discussed. Salman et al. (2015) proposed a deep learning-based weather forecasting model using rich hierarchical weather representations, and the models are tested using large volumes of weather data. In this paper, mainly three deep learning models for weather forecasting, namely RNN, CRBM, and CNN, are described. Each of these models is trained and tested using the predetermined weather dataset. A parameter learning algorithm for each model, Gradient Descent for CRBM, is implemented to gain testing error below the predetermined threshold value. Model performance is evaluated using a k-fold cross-validation technique in which the weather dataset is divided randomly into k folds. Kaba et al. (2018) developed a deep neural network-based daily GSR prediction model for various stations in Turkey. Flores-Vergara et al. (2019) proposed a deep learning architecture for predicting ozone pollution at Santiago in South America.

A new deep network layer model called “Dynamic Convolutional Layer” for short-range weather prediction is described by Klein et al. (2015). Three different datasets validate the effectiveness of the model. The dynamic convolution layer has two inputs, namely feature maps and filters. The whole system is trained using the back-propagation algorithm. The open-source Caffe library is used to implement the Dynamic Convolutional Layer. All the weights in the deep neural network are initialized according to the Xavier initialization. Li et al. (2016) proposed a novel spatio-temporal deep learning-based air quality prediction method using stacked autoencoder to extract inherent air quality features. The authors then compared the performance of this model with artificial neural networks, autoregression moving average, and support vector regression models. The empirical outcomes illustrate that the proposed method for performing air quality predictions has a superior performance. The efficiency of the suggested model is evaluated using the error indicators RMSE, MAE, and MAPE. A deep learning-based architecture for the prediction of the accumulated daily precipitation for the next day is suggested by Hernandez et al. (2016). The model utilized an autoencoder for reducing and

capturing non-linear relationships between attributes and an MLP for the prediction task.

Ghoneim and Manjunatha (2017) presented a new deep learning-based ozone level prediction model. Different deep architectures have experimented, and architecture with ten hidden layers, 120 hidden neurons in each hidden layer, and 500 epochs is found to be the best configuration. The proposed model predicted the ozone level more accurately. Error-values MSE, MAE, and RMSE of the proposed model are determined to be minimal compared to conventional machine learning models like SVM, ANN, and GLM. Gensler et al. (2016) presented a deep learning-based architecture for solar power forecasting. The proposed architecture utilized stacked autoencoder and LSTM for forecasting of solar power. Hosain et al. (2015) proposed a deep neural network model with Stacked Denoising Autoencoders for predicting air temperature. Experimental results revealed that stacked denoising autoencoders performed better than the neural network in temperature prediction as it produced high accuracy.

A novel approach for precipitation forecasting based on deep belief nets, called Deep Belief Network for Precipitation Forecast, is proposed by Zhang et al. (2017), and the characteristics hidden in hydrological time series data are easily extracted. The proposed deep learning-based model is compared with other traditional machine learning approaches. The experimental results show that the DBNPF model is more efficient and effective compared to the benchmark models such as the RBF neural network, SVM, ARIMA, ELM, and sparse autoencoder. A wind direction prediction system is proposed by Hou et al. (2019) using Z-score normalization and LSTM networks. Liang et al. (2018) recommended a multivariate wind speed forecasting system using a stacked LSTM network for a wind station in West Texas, USA. Zaytar and Amrani (2016) proposed a weather forecasting model using the Stacked LSTM network, and the performance of the model is estimated using weather data collected from nine cities. The studies verify that deep learning architectures like RNN and LSTM have strong self-learning capability and are better models for time series prediction. Table 3 provides the parameters and the metrics used by the deep learning predictors.

5.1.3. Hybrid models

Hybrid models are devised by combining two or more models, where linear or nonlinear models can be combined. These are the most recent models which yield better performance. Analysis of studies confirms that hybrid models deliver better forecasting accuracy than individual models. Khashei and Bijari (2011) proposed a hybrid system combining linear model ARIMA and nonlinear model ANN and found them more effective as it produced higher forecasting accuracy. Khodayar and Teshnehlab et al. (2015) proposed a short-term wind speed forecasting model using stacked autoencoder for Colorado, USA. A rough regression layer is trained on the top of the SAE to forecast wind speed efficiently. The rough regression layer is employed to handle the uncertainties present in wind speed time series. Su et al. (2018) developed an ultra-short-term wind power forecasting model based on data decomposition techniques and Elman neural network.

Yu et al. (2017) proposed a prediction system to forecast wind speed using a wavelet transform and Elman neural network, where the wavelet transform is employed to the input time series to denoise the signal. Saba et al. (2017) recommended a hybrid model using MLP and RBF to enhance the efficiency of weather prediction. The accuracy is assessed using the Correlation coefficient, RMSE, and scatter index. The proposed hybrid model has better forecast accuracy compared to single MLP and RBF networks. Liu et al. (2013) proposed a wind speed forecasting system by combining Wavelet, Wavelet Packet, Time series, and ANN. The proposed model is compared with neuro-fuzzy, ANFIS, wavelet packet-RBF,

Table 3

Summary of the parameters used by Deep learning models.

Algorithm Used	Reference	Parameters used									Metrics used				
		Temperature	Wind speed	Wind direction	Air pressure	Humidity	Precipitation	Rainfall	Dew point	Ozone	R ²	MSE	RMSE	MAE	MAPE
RNN	(Salman et al., 2015)	✓	✓		✓	✓									
RNN	(Flores-Vergara et al., 2019)									✓			✓		
DNN	(Hernández et al., 2016)	✓	✓	✓	✓	✓	✓					✓	✓		
DNN	(Ghoneim and Manjunatha, 2017)	✓	✓	✓	✓	✓			✓	✓		✓	✓	✓	
SDAE	(Hossain et al., 2015)	✓	✓		✓	✓							✓	✓	
DBNPF	(Zhang et al., 2017)	✓	✓	✓	✓	✓	✓						✓	✓	
LSTM	(Hou et al., 2019)		✓											✓	✓
Stacked LSTM	(Liang et al., 2018)		✓								✓	✓	✓	✓	✓
Stacked LSTM	(Zaytar and El Amrani, 2016)	✓													

Table 4

Summary of the parameters used by Hybrid models.

Algorithm Used	Reference	Parameters used									Metrics used				
		Temperature	Wind speed	Wind direction	Air pressure	Humidity	Precipitation	Rainfall	Dew point	Ozone	R ²	MSE	RMSE	MAE	MAPE
SAE-RoughR	(Khodayar and Teshnehlab, 2015)	✓											✓		
EMD-ENN	(Su et al., 2018)		✓										✓	✓	✓
Wavelet-ENN	(Yu et al., 2017)		✓										✓	✓	✓
MLP-RBF	(Saba et al., 2017)	✓	✓				✓	✓	✓		✓		✓		
Wavelet Packet-ANN	(Liu et al., 2013)		✓									✓		✓	✓
SAE-DNN-SVR	(Liu et al., 2015)	✓	✓	✓					✓		✓				
SAE-DBN	(Sergio and Ludermir, 2015)		✓									✓		✓	✓
SAE-LSTM	(Wu et al., 2019)	✓	✓										✓	✓	✓
Wavelet-ELM	(Mi et al., 2017)		✓										✓	✓	✓
HDT-BPNN	(Qu et al., 2019)		✓										✓	✓	✓
ACO-ELM	(Carrillo et al., 2017)		✓								✓			✓	✓
Wavelet-SVR	(Dhiman et al., 2019)		✓										✓	✓	✓
HMD – OSORELM	(Zhang et al., 2019)		✓										✓	✓	✓
SSA-EMD-SVM	(Mi et al., 2019)		✓										✓	✓	✓
GREY – ELM	(Qolipour et al., 2019)		✓									✓	✓		✓
ELM-ENN-LSTM	(Chen et al., 2019)		✓											✓	✓
MMODA	(Liu et al., 2020)		✓										✓	✓	✓

and persistent model. The results demonstrated that the wavelet packet ANN model outperformed all other models. Liu et al. (2015) developed a model using deep neural networks to predict the weather, especially temperature and wind speed, accurately for the next 24 h using big weather datasets. The weather parameter is predicted using stacked autoencoder DNN architecture with SVR trained on top of it. Sergio and Ludermir (2015) suggested a deep learning-based model to forecast mean wind speed in the North-eastern region of Brazil using SAE and DBN. Wu et al. (2019) developed a DNN based model for wind speed forecasting using stacked autoencoder and LSTM, where Stacked Autoencoder is selected to extract in-depth features from the weather datasets for improving forecasting accuracy further. NMSE and NMAE are used to evaluate the prediction performance of the proposed

model. Mi et al. (2017) defined a new hybrid method for wind speed forecasting using WT, WPD, ELM, and Outlier correction algorithm, where the Wavelet Transform is applied to denoise the input signal. The WPD algorithm is then employed to partition the input wind speed series.

Qu et al. (2019) built a hybrid system to predict the wind speed based on HDT and a backpropagation neural network optimized using the flower pollination algorithm. Carrillo et al. (2017) proposed a hybrid predictive model based on ACO and ELM to forecast wind power. The datasets collected from two wind farms in Spain are utilized to estimate the efficiency of the model. A novel hybrid weather forecasting model using a multi-task agent combined with supervised and unsupervised learning is proposed by Arunachalam et al. (2015). Dhiman et al. (2019) suggested a wind speed predic-

Table 5

Classification of models based on the parameter to be predicted.

Sl No.	Parameter predicted	References
1	Temperature	(Rasp and Lerch, 2018; Nayak et al., 2012; Liu et al., 2018, 2015; Abhishek et al., 2012; Khajure and Mohod, 2016; Suksri and Kimpan, 2016; Narvekar and Fargose, 2015; Yu et al., 2016; Rasel et al., 2017; Singh et al., 2019; Ahmed, 2015; Mohammadi et al., 2015; Ismail et al., 2016; Jayanthi and Sumathi, 2017; Suryanarayana et al., 2019; Hossain et al., 2015)
2	Wind speed	(Tarade and Katti, 2011; Kulkarni et al., 2008; Kavasseri and Seetharaman, 2009; Erdem and Shi, 2011; Liu et al., 2011, 2012, 2013, 2015, 2020; Chen et al., 2009, 2019; Cadenas et al., 2016; Filik and Filik, 2017; Hou et al., 2019; Liang et al., 2018; Khodayar and Teshnehlab, 2015; Su et al., 2018; Yu et al., 2017; Sergio and Ludermer, 2015; Wu et al., 2019; Mi et al., 2017, 2019; Qu et al., 2019; Carrillo et al., 2017; Dhiman et al., 2019; Zhang et al., 2019; Qolipour et al., 2019)
3	Rainfall	(Reddy and Babu, 2017; Tran Anh et al., 2019; Zaw and Naing, 2009; Suhartono et al., 2012; Hung et al., 2009; Kashiwao et al., 2017; Mislan et al., 2015; Narvekar and Fargose, 2015; Rasel et al., 2017; Lu and Wang, 2011; Shabariram et al., 2016; Ahmed, 2015; Navadia et al., 2017; Hernández et al., 2016; Zhang et al., 2017; Saba et al., 2017)
4	Precipitation	(Du et al., 2017; Hernández et al., 2016; Zhang et al., 2017)
5	Pressure	(Khajure and Mohod, 2016; Suryanarayana et al., 2019)
6	Dew point	(Khajure and Mohod, 2016; Hasan et al., 2016; Mohammadi et al., 2015; Saba et al., 2017)
7	Ozone level	(Flores-Vergara et al., 2019; Ghoneim and Manjunatha, 2017)

tion system by combining the Wavelet Transform and SVR. Zhang et al. (2019) proposed a wind speed prediction model based on the Hybrid Mode Decomposition method and OSORELM, where HMD is employed to decompose the input wind series. HMD is designed by combining Variational Mode decomposition, Sample Entropy, and Wavelet Packet Decomposition. Crisscross optimization algorithm is applied to optimize the parameters of the OSORELM model. The simulation results demonstrate that the performance of the proposed hybrid model has substantially improved. Mi et al. (2019) described an improved hybrid wind speed forecasting model based on SSA, EMD, and convolutional SVM. Qolipour et al. (2019) recommended a hybrid wind speed forecasting system using Grey ELM and the efficiency of the model is assessed by using four different datasets. By combining the features of ELM, Elman neural network, and LSTM networks, Chen et al. (2019) suggested a hybrid short-term wind speed forecasting model. Liu et al. (2020) recommended a wind speed forecasting model by combining data pre-treatment strategy, modified multi-objective optimization (MMO) algorithm, and various forecasting models. A pre-treatment strategy is applied to capture the significant features of wind speed data, and a modified MMO algorithm is employed to compute the optimal weights of the suggested model. The performance of the model is compared with ARIMA, BPNN, ELM, Elman neural network, and General Regression Neural Network models.

The summary of the parameters and the metrics used by various hybrid weather forecasting models is listed in Table 4.

5.2. Classification based on parameter to be predicted

Forecasting models can be categorized as temperature prediction models, wind speed prediction models, rainfall prediction models, dew point prediction models, and so on depending upon the parameter to be predicted. Each prediction model forecasts a particular parameter based on previous observations. Temperature forecasting models aim at predicting the minimum, maximum, or average temperature of a location, based on various weather parameters. The maximum temperature forecast depends on sunshine during day time, whereas the minimum temperature forecast relies on cloud conditions at night.

Temperature prediction plays a vital role in agriculture. Prior information about weather helps farmers to take necessary decisions to enrich their crop cultivation. The extreme temperature will cause severe damages to plants and animals, and it is always a great concern for farmers. The temperature should be optimal for the growth of plants. Low temperatures may cause injury to crops. So, the prediction based on temperature is essential in agriculture.

Likewise, wind speed prediction models are intended for forecasting wind speed. Wind speed prediction is an important activity as the wind has applications in various domains such as power generation, agriculture, Industry, Naval Systems, and the marine world. The wind is the process of moving air from high pressure to low pressure, and wind speed is calculated by taking into account the difference between them. As the difference increases, the wind speed also increases.

Quantitative rainfall is the amount of rainfall received over a period of time in a given area. Rain forecasting is essential for the agricultural industry and the flood monitoring system. Precipitation is frozen water or liquid that forms in the atmosphere and drops to the Earth. This weather parameter will help farmers to make decisions on activities like irrigation, spraying, and harvesting.

Atmospheric air pressure is the weight exerted by the air on the surface of the Earth, and it varies with altitude. Atmospheric pressure decreases with increasing altitude. Dew point specifies how much water is present in the air, and it will get a high value when there is more water vapor in the air. Dew point is essential for the growth of desert plants and also for plants growing in areas with moderate rainfall. The quantity of moisture contained in the air of the lower atmosphere is termed as humidity, which is essential for leaf growth, photosynthesis, pollination, and economic yield. Ozone is one of the hazardous pollutants in the lower atmosphere. The concentration of ozone in the atmosphere discloses its impact on plants, humans, and other organisms. Predicting and controlling ozone concentrations can reduce the effects of tropospheric ozone on human health and the environment.

Table 5 provides details of some of the studies conducted by various researchers for forecasting temperature, wind speed, dew point, rainfall, precipitation, and pressure. Several studies have been proposed by various researchers, as shown in Table 5.

5.3. Discussion

From the analysis of various weather prediction models, it has been observed that hybrid models are proved to be more reliable models for weather forecasting. Most of the wind speed forecasting models discussed in the literature are univariate models, as the forecasting involves only a single variable. Fig. 8 explains the details of various prediction models reviewed in this study. Fig. 9 illustrates the summary of the quantitative indicators used by various models presented in the literature. Fig. 10 depicts the percentage of quantitative indicators used by various models. It can be comprehended from the figure that most of the prediction models are wind speed prediction models, which are followed by temperature prediction models.

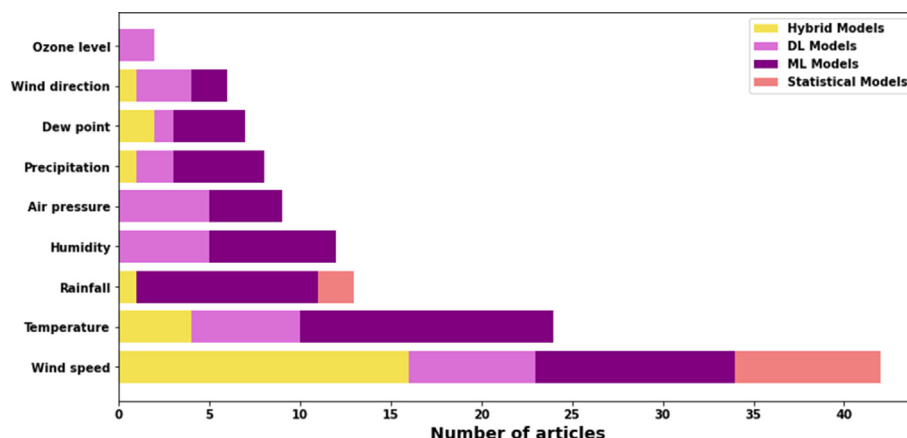


Fig. 8. Details of various forecasting models based on parameters.

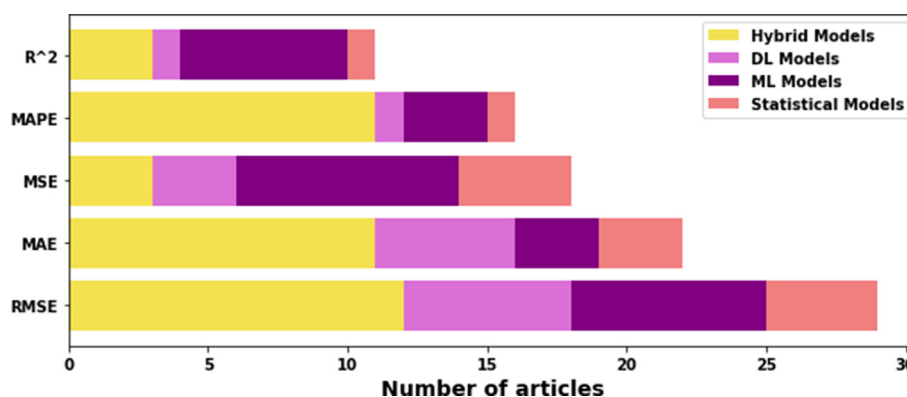


Fig. 9. Details of quantitative indicators used by various forecasting models.

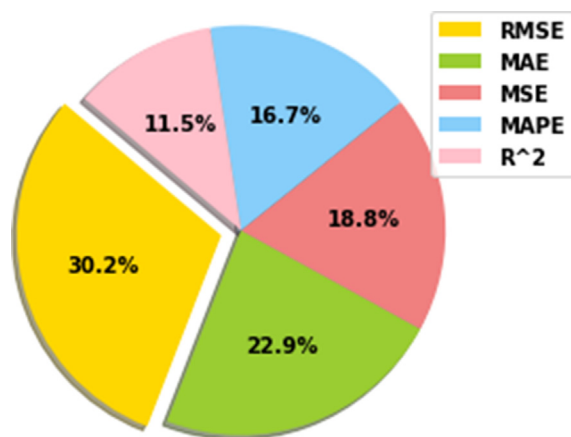


Fig. 10. Percentage of quantitative indicators used.

Moreover, about 30% of the forecasting models employed RMSE as the quantitative indicator for assessing the performance of the forecasting models, as illustrated in Fig. 10. RMSE is established to be a commonly used metric for validating the efficacy of the models. RMSE is very easy to differentiate, making it easy to use in derivative based methods. Since the errors are squared before they are averaged, the RMSE places a relatively high weight on large errors. Moreover, absolute values are not used by RMSE, and hence, gradient and distance can be measured more effectively. This might be the reason for using squared errors such as RMSE in prediction models.

6. Results and analysis

6.1. Open weather datasets

Some of the publicly available datasets for experimental analysis are detailed in this section. Some researchers publish their data online along with journals as supplementary files or in some public archives such as GitHub so that others can use these datasets for their research. This type of online datasets promotes reproducible research.

- 1) *US Weather Events* is an open weather dataset that contains information of about five million events. The weather parameters include rain, storm, snow, and freezing conditions of 49 states of the US from 2016 to 19 (Moosavi et al., 2019).
- 2) *Historical Hourly Weather Data* is also an open dataset that contains hourly data of temperature, humidity, air pressure for five years from 2012 to 17. The dataset includes data from 30 US and Canadian Cities, as well as 6 Israeli cities, and the dataset can be downloaded from the OpenWeather-Map website using Weather API under the ODbI license.
- 3) *Rain in Australia* is a publicly available rain dataset that contains weather observations from weather stations of Australia, which can be used for rainfall prediction. The dataset can be downloaded from <http://www.bom.gov.au/climate/data>.
- 4) *Daily global surface weather observations* from more than 9000 weather stations from 1929 to present are available from NOAA and include GSOD data acquired from the ISH

dataset. The USAF Climatology Center provides the ISH dataset. The weather parameters include temperature, wind speed, dew point, visibility, and pressure.

- 5) *Historical Daily Weather Data* includes weather data from 163 countries from 1st January 2020 to 21st April 2020. The countries are chosen based on Johns Hopkins COVID-19 dataset. The dataset provides daily information on temperature, pressure, humidity, ozone levels, visibility, and precipitation. This data can be extracted using the Dark Sky API. The goal of this dataset is to determine the impact of weather elements on Covid19 transmission and to propose forecast models based on these factors.
- 6) *Weather Istanbul dataset* contains parameters such as rainfall, average wind speed, average humidity, and average pressure. The dataset can be downloaded from the website <https://www.worldweatheronline.com>.
- 7) *Weather Underground* provides weather information such as temperature, humidity, wind speed, pressure, and visibility from major cities across the world. The dataset can be downloaded from Weather Underground, available at www.wunderground.com.
- 8) *In-situ weather data* provides atmospheric observations from aircraft and surface stations. The collected data may not be of good quality, and hence some quality control and preprocessing steps to be carried out before processing the data.

6.2. Evaluation criteria

Evaluation plays a vital role in any forecasting model. The accuracy of the forecasting models can be adequately assessed using

the metrics, MAE (Mean Absolute Error), MSE (Mean Squared Error), RMSE (Root Mean Squared Error), MAPE (Mean Absolute Percentage Error), and R^2 (Coefficient of Determination). Various models employ different metrics for the evaluation of the models. MAE is demarcated as the absolute difference between the predicted value ($\hat{y}$) and the actual value(y). RMSE is described as the square root of the average of squared differences between predicted value ($\hat{y}$) and the actual value(y). MAPE is the relative percentage error between actual and predicted values. The lower the error values, the higher will be the prediction accuracy of the corresponding model. R^2 , the Coefficient of Determination, is a statistical indicator that evaluates the goodness of fit of the model, which varies from 0 to 100. Higher R^2 values represent smaller differences between the actual and the predicted values. $R^2 = 100$ means the prediction model performs exceptionally well on the testing datasets. These error indicators are defined using equations (25) to (29). Among the quantitative indicators employed for assessing the performance of the models, MAE, MSE, RMSE, and MAPE are negative indicators, which means that the lower values of which indicate better prediction accuracy.

$$MAE = \frac{1}{n} \sum_{j=1}^n |y_j - \hat{y}_j| \quad (25)$$

$$MSE = \frac{1}{n} \sum_{j=1}^n (y_j - \hat{y}_j)^2 \quad (26)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{j=1}^n (y_j - \hat{y}_j)^2} \quad (27)$$

Table 6

Error values of Temperature prediction models.

References	MSE	RMSE	MAE	MAPE	R^2
(Nayak et al., 2012)				4.4130	
(Liu et al., 2018)	1.0600	1.0296			99.6
(Abhishek et al., 2012)	2.7100	1.6460			
(Sukri and Kimpan, 2016)	0.0212	0.1476			
(Yu et al., 2016)	0.0281	0.1676	0.1062	0.0279	
(Rasel et al., 2017)	10.956	3.3100	2.2900		
(Ahmed, 2015)					93.36
(Mohammadi et al., 2015)	0.4501	0.6709		0.5203	98.77
(Hossain et al., 2015)	1.9044	1.3800			
(Liu et al., 2015)					91.50

Table 7

Error values of Wind speed prediction models.

References	MSE	RMSE	MAE	MAPE	R^2
(Tarade and Katti, 2011)	1.4328	1.1969			
(Kulkarni et al., 2008)	3.21126	1.79200			
(Kavasseri and Seetharaman, 2009)	28.6225	5.35			
(Liu et al., 2012)	0.2322	0.48187	0.1745	0.87	
(Cadenas et al., 2016)	1.51	1.22882	0.91		
(Filik and Filik, 2017)	0.42354	0.65080	0.5046		
(Liang et al., 2018)	0.00297	0.05448	0.0428		99.917
(Khodayar and Teshnehlab, 2015)	0.00281	0.05300			
(Yu et al., 2017)	0.22090	0.47000	0.3593	8.7200	
(Liu et al., 2013)	0.54330	0.73709	0.4396	3.5500	
(Liu et al., 2015)					87.100
(Sergio and Ludermir, 2015)	0.59920	0.77408	0.5962	0.1120	
(Mi et al., 2017)	0.08280	0.28770	0.2353	1.2500	
(Qu et al., 2019)	0.00434	0.06590	0.0533	1.0047	
(Dhiman et al., 2019)	2.01640	1.42000			
(Zhang et al., 2019)	0.00025	0.01590	0.0125	0.1294	
(Mi et al., 2019)	0.16000	0.40000	0.3100	5.3500	
(Qolipour et al., 2019)	0.00099	0.03142		0.03784	99.376
(Chen et al., 2019)			0.4363	6.7007	
(Liu et al., 2020)	0.04260	0.20640	0.1659	3.1539	

Table 8

Error values of Rainfall prediction models.

References	MSE	RMSE	MAE	R ²
(Suhartono et al., 2012)	144.5766	12.024		
(Hung et al., 2009)	0.5041	0.710		94.00
(Mislan et al., 2015)	0.7010	0.8373		
(Rasel et al., 2017)	413.308	20.330	0.95	
(Hernández et al., 2016)	40.110	6.330		
(Zhang et al., 2017)	5.09404	2.257	1.632	
(Saba et al., 2017)	146.000	12.083		95.00

Table 9

Error values of Precipitation prediction models.

References	MSE	RMSE	MAE
(Du et al., 2017)	0.231	0.480	
(Hernández et al., 2016)	40.110	6.330	
(Zhang et al., 2017)	0.667	0.817	0.494

$$MAPE = \frac{1}{n} \sum_{j=1}^n \left| \frac{y_j - \hat{y}_j}{y_j} \right| \quad (28)$$

$$R^2 = 1 - \frac{\sum_{j=1}^n (y_j - \hat{y}_j)^2}{\sum_{j=1}^n (y_j - \bar{y}_j)^2} \quad (29)$$

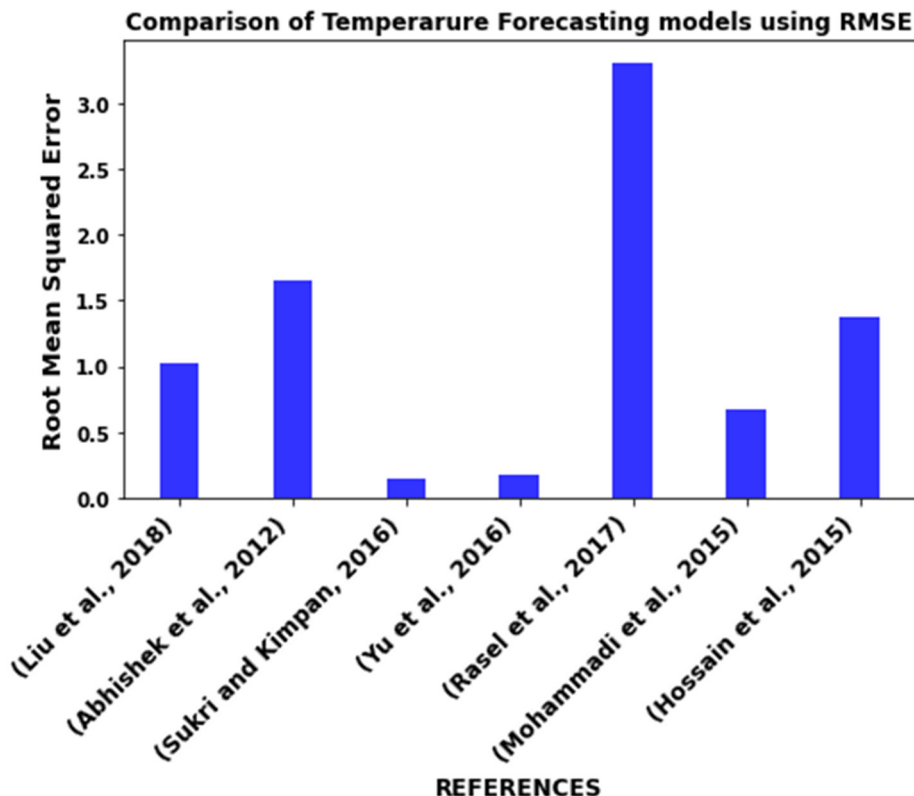
6.3. Analysis of forecasting models

A huge amount of research has performed in weather forecasting, especially in temperature and wind speed forecasting, with drastic improvements in the performance of the models. Best reported error values, namely MAE, MSE, RMSE, MAPE, and R² on various models, are presented in the tables below. Readers are

requested not to deduce a testimonial regarding the performance of weather forecasting models by merely looking at the tables and figures shown below because these reported error values might be strongly impacted by the forecasting duration, the number of variables involved, and the methodology employed in different methods. Moreover, according to the variations in the data format and size of datasets used in the various models, the results obtained may also tend to vary. Hence, it cannot be concluded that some methods are more accurate than other methods and vice versa. However, the results are plotted in the figures only for analysis purposes.

Tables 6–9 list the best-reported error values of different weather forecasting models. Table 6 illustrates the error measures of temperature prediction models presented in the literature. The error measures of wind speed, rainfall, and precipitation forecasting models are described in Tables 7, 8, and 9, respectively.

Even though various researchers employed MSE, RMSE, MAE, MAPE, and R² for assessing the effectiveness of their models, RMSE has been selected for comparing the models in this study. Fig. 11 shows the error measures of various temperature forecasting models discussed in the literature in terms of RMSE. The models proposed by Suksri and Kimpan (2016) and Yu et al. (2016) reported lower RMSE values and hence exhibit better prediction accuracy

**Fig. 11.** Comparison of Temperature Forecasting models based on RMSE.

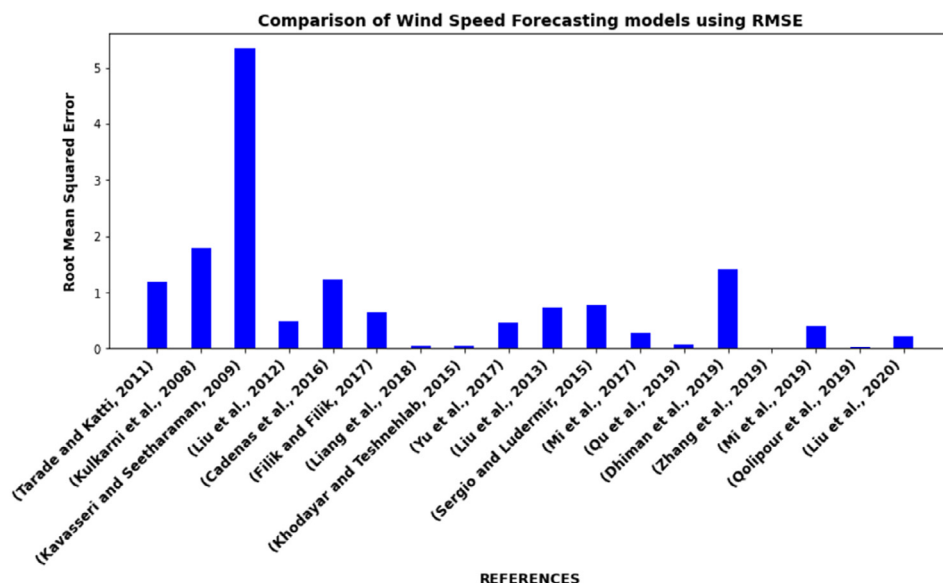


Fig. 12. Comparison of Wind Speed Forecasting Models based on RMSE.

in temperature forecasting. It can be understood that ANN and SVM based temperature prediction models exhibit better performance.

The prediction accuracy of wind speed forecasting models based on RMSE is illustrated in Fig. 12. It can be perceived that the models proposed by Liang et al. (2018), Khodayar and Teshnehlab (2015), Qu et al. (2019), Zhang et al. (2019), and Qolipour et al. (2019) reported minimum error values in wind speed forecasting compared to other models. It has been observed that hybrid architectures yield better prediction accuracy compared to individual models. Hybrid models outperform other models, specifically in wind speed forecasting.

The comparison of rainfall and precipitation forecasting models based on RMSE are provided in Figs. 13 and 14, respectively. It can be observed that both machine learning and deep learning-based models perform well in rainfall and precipitation predictions. Hung et al. (2009), Mislan et al. (2015), and Zhang et al. (2017) reported the best prediction accuracy in rainfall forecasting. The models developed by Du et al. (2017) and Zhang et al. (2017) reported minimum error values in precipitation forecasting.

The detailed survey concludes that Artificial Neural Networks and Support Vector machines are the most commonly used and reliable machine learning architectures for weather prediction. Nonlinear problems that are difficult to solve using traditional techniques can be solved using these architectures. ANNs can be employed for weather forecasting because of the ability to find relationships between variables within the datasets. Deep learning is valued as a promising technique for weather forecasting. The studies confirm that deep learning and hybrid models are reliable for accurate weather forecasting, especially wind speed forecasting. Proper and accurate weather forecasting will help people for better decision making and improve safety across all aspects of human life.

7. Challenges and future directions

From the analysis of the works presented in the literature, the following challenges have been noticed. The efforts which are taken to overcome these challenges can be considered as potential future directions.

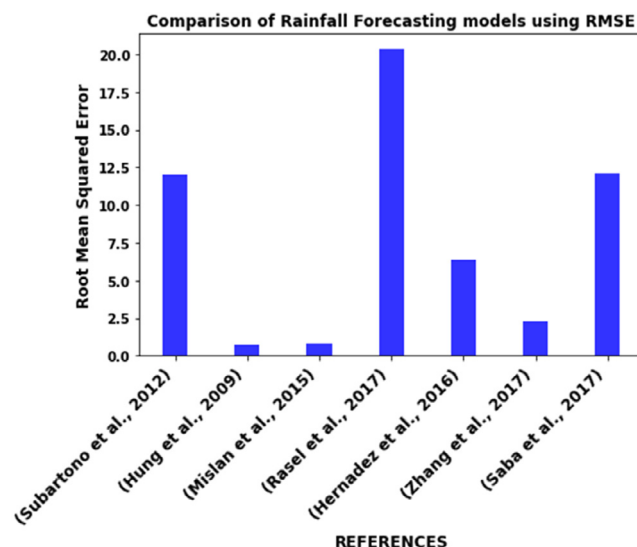


Fig. 13. Comparison of Rainfall Forecasting Models based on RMSE.

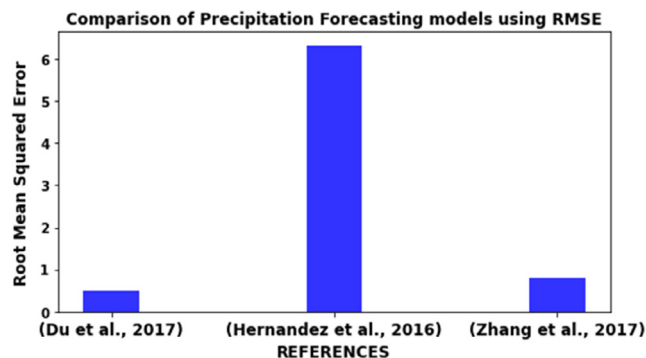


Fig. 14. Comparison of Precipitation Forecasting Models based on RMSE.

- (1) Data collection is an important step in any forecasting problems. The availability and accessibility of real-time and good quality weather datasets is still a challenging task.
- (2) It can be noticed that most of the works presented in the literature utilized small datasets for experiments. So, more works on massive datasets using different techniques need to be addressed.
- (3) A large number of works have been performed based on various machine learning, deep learning, and hybrid techniques for wind speed forecasting. However, works related to other weather parameters are comparatively less. Hence, measures are to be taken to focus more on rainfall and dew point prediction models using various novel techniques as a future research direction.
- (4) It is observed that most of the studies presented in the literature make use of structured data sets for storing weather information. Efforts are to be made for developing forecasting models using atmospheric images and can be considered as prospective future research concerns.
- (5) Most of the wind speed forecasting models presented in the literature are based on univariate LSTM networks, and correlated features are not included in the input features. Hence, more multivariate models need to be developed in the future by incorporating correlated features such as temperature and wind direction into the input features. The effectiveness of such models should also be explored as well.
- (6) It can be observed that the performance indicators applied in the articles reviewed include an assessment of the prediction accuracy only. Some measures need to be addressed to evaluate the stability to ensure that the proposed model can be used to predict more types of data, which can be considered as a future research concern.

8. Conclusion

With the advancement of Big Data technologies and deep learning techniques, weather forecasting and climate prediction can be done effectively and accurately. The proposed survey discusses the recent research works related to weather forecasting, along with a detailed analysis of the results. The classification of weather forecasting models are done mainly based on methodology employed and weather parameter to be predicted. The main limitation identified in the evaluation of the existing systems is the lack of stability assessment of the weather forecasting models. All the existing models only evaluate the prediction accuracy. The results claimed by the authors are analyzed for assessing the performance of different techniques. ANN and SVM are established to be more reliable machine learning techniques for weather forecasting, and recently, neural networks with deep architectures and hybrid models offer promising results in the field of weather prediction. The survey provides the state of the art models for weather forecasting, its challenges, available open datasets, and future research directions. This detailed literature review will help researchers who intend to explore the field of weather forecasting as a reference guide.

Declaration of Competing Interest

The authors declared that there is no conflict of interest.

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